

NU CSE 1st Semester Calculus Questions (2009-2019)

Book name :CSE Kit NU

Calculus | 373

DIFFERENTIAL CALCULUS CHAPTER 1 FUNCTION

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1.	What is calculus?	
2.	Define function.	(2018) (IST-18, DCC-18)
3.	Define algebraic function.	(2018) (IST-18, DCC-18)
4.	Define domain and range.	(2019, 2017, 2015) (DCC-18)
5.	Draw the graph of the function ; $f(x) = x + 1 + x + x - 1 $	(DCC-18)
6.	A function is defined as follows :- $f(x) = \begin{cases} x^2 & \text{when } x < 0 \\ x & \text{when } 0 \leq x \leq 1 \\ \frac{1}{x} & \text{when } x > 1 \end{cases}$ Find the domain and range of the function	(2015)
7.	Determine the domain and range of the following function $f(x) = \begin{cases} x^2 + 1, & x < 0 \\ x, & 0 \leq x \leq 1 \\ \frac{1}{x}, & 1 < x \end{cases}$	(2017, 2012)
8.	Draw the graph of the function define by $f(x) = x - 1 + x + 1 $	
9.	Draw the graph of the function defined by $f(x) = \begin{cases} 1 + x & \text{when } -1 \leq x < 0 \\ 1 - x & \text{when } 0 \leq x < 1 \\ 0 & \text{when } 1 < x \end{cases}$ Determine the range and domain of the function.	(2010)
10.	Definition: Concave up, concave down and inflection point.	(2017), (2018)

CHAPTER 2

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LIMIT

1.	Define limit of a function.	(2018)
2.	By the definition of (δ, ϵ) , prove that $\lim_{x \rightarrow 5} \frac{x^2 - 25}{x - 5} = 10$	(2017)
3.	By the definition of (δ, ϵ) , prove that $\lim_{x \rightarrow 5} \frac{x^2 - 16}{x - 4} = 6$	(2018, 2017, 2013)
4.	By the definition of (δ, ϵ) , prove that $\lim_{x \rightarrow 4} (2x - 2) = 6.$	(IST)
5.	By the definition of (δ, ϵ) , prove that $\lim_{x \rightarrow 3} \frac{2x^2 - 18}{x - 3} = 12$	(2016)
6.	By the definition of (δ, ϵ) , prove that $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3} = 6$	(2012)
7.	By the definition of (δ, ϵ) , prove that $\lim_{x \rightarrow 2} \frac{2x^2 - 8}{x - 2} = 8$	(2015, 2011)
8.	A function is defined as follows :- $f(x) = \begin{cases} \tan\left(\frac{x}{2}\right) & \text{when } x < \frac{\pi}{2} \\ 3 - \frac{\pi}{2} & \text{when } x = \frac{\pi}{2} \\ \frac{x^3 - \pi^{\frac{3}{8}}}{x - \frac{\pi}{2}} & \text{when } x > \frac{\pi}{2} \end{cases}$ Find $\lim_{x \rightarrow \frac{\pi}{2}} f(x)$.	
9.	1. Given $f(x) = \begin{cases} 1 + 2x, & \frac{1}{2} \leq x < 0 \\ 1 - 2x, & 0 \leq x < \frac{1}{2} \\ -1 + 2x, & x > \frac{1}{2} \end{cases}$ Show that, $\lim_{x \rightarrow 0} f(x)$ and $\lim_{x \rightarrow \frac{1}{2}} f(x)$ exists	[2014]

CHAPTER 3 **PAGE No: 401**
CONTINUITY AND DIFFERENTIABILITY

1.	Define Continuity And Differentiability.	(2014,2013,2011)
2.	A function is defined as follows :- $f(x) = \begin{cases} 5x - 4; & \text{when } 0 < x \leq 1 \\ 4x^2 - 3x; & \text{when } 1 < x < 2 \end{cases}$ Discuss the continuity and differentiability of the function $f(x)$ at $x = 1$. or not.	(2017,2011)
3.	A function is defined as follows :- $f(x) = \begin{cases} 3 + 2x & \text{when } -\frac{3}{2} \leq x < 0 \\ 3 - 2x & \text{when } 0 \leq x < \frac{3}{2} \\ -3 - 2x & \text{when } \frac{3}{2} \leq x \end{cases}$ Discuss the continuity and differentiability of the function $f(x)$ at $x = 0$ and $x = \frac{3}{2}$.	(2017,2016,2012) (2018)
4.	A function is defined as follows :- $f(x) = \begin{cases} 1 & \text{when } x < 0 \\ 1 + \sin x & \text{when } 0 \leq x < \frac{1}{2}\pi \\ 2 + \left(x - \frac{1}{2}\pi\right)^2 & \text{when } x \geq \frac{1}{2}\pi \end{cases}$ discuss the continuity and differentiability of the function $f(x)$ at $x = \frac{\pi}{2}$	(2015)
5.	A function is defined as follows :- $f(x) = \begin{cases} -x & \text{when } x < 0 \\ x & \text{when } 0 \leq x \leq 1 \\ 2 - x & \text{when } x > 1 \end{cases}$ Discuss the continuity and differentiability of the function $f(x)$ at $x = 0$ and $x = 1$.	(2013)
6.	A function is defined as follows :- $f(x) = \begin{cases} -x & \text{when } -2 \leq x \leq 0 \\ x & \text{when } 0 < x < 1 \\ 3 - x & \text{when } 1 \leq x \leq 2 \end{cases}$ Discuss the continuity and differentiability of the function $f(x)$ at $x = 0$ and $x = 1$.	(2014)
7.	1. A function is defined as follows :-	(2011)

	$f(x) = \begin{cases} x^2 & \text{when } x < 0 \\ x & \text{when } 0 \leq x \leq 1 \\ \frac{1}{x} & \text{when } x > 1 \end{cases}$ Test the continuity and differentiability of the function $f(x)$ at $x = 0$ and $x = 1$. Also draw the graph.	
8.	A function is defined as follows :- $f(x) = \begin{cases} 1+x; & \text{when } -1 \leq x < 0 \\ 1-x; & \text{when } 0 \leq x \leq 1 \end{cases}$ Discuss the continuity and differentiability of the function $f(x)$ at $x = 0$.	2007
9.	A function is defined as follows :- $f(x) = \begin{cases} 5x-4; & \text{when } 0 < x \leq 1 \\ 4x^2-3x; & \text{when } 1 < x < 2 \end{cases}$	
10.	A function defined as follows: $f(x) = \begin{cases} 5x-4 & \text{when } 0 < x \leq 1 \\ 4x^2-3x & \text{when } 1 < x < 2 \\ 3x+4 & \text{when } x \geq 2 \end{cases}$ Test the continuity at $x=1$ and $x=2$	(2017,2019)
11.	A function $f(x)$ is defined as follows: $f(x) = \begin{cases} x \sin\left(\frac{1}{x}\right), & -1 < x < 1, x \neq 0 \\ 0, & x = 0 \end{cases}$	

CHAPTER 4 DIFFERENTIATIONS

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	Find $\frac{dy}{dx}$ of:	
1.	$y = \sin(x+y);$	(2013)
2.	$x^y + y^x = a^b;$	(2013)(DCC-18)
3.	$y = \tan^{-1} \sqrt{\frac{1-\cos x}{1+\cos x}}$	(2018,2013)
4.	$y = \log_a x + \log_x a;$	(2018, 2016, 2010)
5.	$y = x^x + x^{\frac{1}{x}}$	(2010)
6.	Differentiate $x^{\sin^{-1} x}$ with respect to $\sin^{-1} x$.	(2013,2010)
7.	Differentiate $x^{\tan^{-1} x}$ with respect to $\tan^{-1} x$.	(2016) (DCC-18)
8.	Differentiate wth respect to x of the following $xy = \sin(x+y)$	
9.	The rate of change of radius of a circle is $\frac{1}{\pi}$. find the rate of change of circumferential length and area of the circle at the instant when the radius is 2 c.m.	

10.	If the area of a circle increases at a uniform rate, show that the rate of increase of perimeter varies inversely as the radius.	
11.	The volume of a spherical balloon is increasing at the rate of 10 cubic centimeter per second. Find the rate of change of its surface at the instant when its radius is 16 centimeters.	
12.	The actual radius of a sphere is 5 cms, but in measurement was .5 mms. Find the error in volume. What is the percentage of error?	
13.	Find $\frac{dy}{dx}$: (i) $y = \tan^{-1}\left(\frac{1+\tan x}{1-\tan x}\right)$ (ii) $y = x \cos^{-1} x + (\sin x) \ln x$ (iii) $x + y = \sin^{-1}\left(\frac{y}{x}\right)$ (iv) $x = a \cos^3 t, y = a \sin^3 t$	[2019]

CHAPTER 5

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SUCCESSIVE DIFFERENTIATION

1.	Define Successive Differentiation	
2.	State and explain Leibnitz's theorem.	(2010, 2011, 2013, 2014, 2015, 2016, 2018)
3.	If $x = \tanh \frac{\ln y}{m}$ then prove that $(x^2 - 1)y_{n+2} + (2n + 1)xy_{n+1} + (n^2 - m^2)y_n = 0$	
4.	If $y = \sin^{-1}(mx)$ then prove that $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} + (m^2 - n^2)y_n = 0$	(2014, 2012) (DCC-18)
5.	If $y = \sinh^{-1}(mx)$ then prove that $(1 + x^2)y_{n+2} + (2n + 1)xy_{n+1} + (n^2 - m^2)y_n = 0$	
6.	If $x = \sinh\left(\frac{\ln y}{m}\right)$ then prove that $(1 + x^2)y_{n+2} + (2n + 1)xy_{n+1} + (n^2 - m^2)y_n = 0$	(2013)
7.	If $x = \sin\left(\frac{\ln y}{m}\right)$ then prove that $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - (n^2 + m^2)y_n = 0$	
8.	If $x = \tanh\left(\frac{\ln y}{m}\right)$ then prove that	

	$(x^2 - 1)y_{n+2} + (2n + 1)xy_{n+1} + (n^2 - m^2)y_n = 0$	
9.	if $y = \sin^{-1}(mx)$ then prove that $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} + (m^2 - n^2)y_n = 0$	
10.	if $y = \sinh^{-1}(mx)$ then prove that $(1 + x^2)y_{n+2} + (2n + 1)xy_{n+1} + (n^2 - m^2)y_n = 0$	

CHAPTER 6

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INDETERMINATE FORMS

	Evaluate	
1.	$\lim_{x \rightarrow 0} \left(\frac{1}{x^2} - \frac{1}{\sin^2 x} \right)$	(2010)
2.	$\lim_{x \rightarrow e} (\ln x)^{\frac{1}{x-e}}$	
3.	$\lim_{x \rightarrow 0} \left(\frac{1}{x^2} - \cot^2 x \right)$	(DCC-18)
4.	$\lim_{x \rightarrow 0} \frac{e^x - e^{-x} - 2x}{x - \sin x}$	(2016, 2014, 2012)
5.	$\lim_{x \rightarrow 0} \frac{3 \tan x - 3x - x^3}{x^5}$	(2015)
6.	$\lim_{x \rightarrow 0} (\sin x)^x$	(2017, 2015)
7.	$\lim_{x \rightarrow 0} \frac{e^x - e^{-x} - 2 \ln(1+x)}{x \sin x}$	(2017)
8.	$\lim_{x \rightarrow 0} \frac{a^x - 1 - x \ln a}{x^2}$	(2017)
9.	Evaluate: $\lim_{x \rightarrow 0} \left(\frac{1}{x^2} - \cot^2 x \right)$	2008

CHAPTER 7

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PARTIAL DIFFERENTIATION

1.	Homogeneous function	(2017)
2.	State and explain Eulers theorem	(2015, 2014, 2012)
3.	If $u = \tan^{-1} \left(\frac{x^3 + y^3}{x+y} \right)$ then show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \sin 2u$.	(2013)
4.	If $u = \tan^{-1} \left(\frac{x+y}{\sqrt{x} + \sqrt{y}} \right)$ then show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + \frac{1}{2} \cot u = 0$.	
5.	If $u = \tan^{-1} \left(\frac{x+y}{\sqrt{x} + \sqrt{y}} \right)$ then show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} =$	(2013)

	$\frac{1}{4} \sin 2u.$	
6.	If $u = \tan^{-1} \left(\frac{x^2+y^2}{x-y} \right)$ then show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} =$ $\frac{1}{2} \sin 2u.$	(2013)
7.	If $u = \sin^{-1} \left(\frac{x+y}{\sqrt{x}+\sqrt{y}} \right)$ then show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{1}{2} \tan u.$	(2014)
8.	If $u = \ln(x^2 + y^2 + z^2)$, then show by Euler's theorem that, $x \left(\frac{\partial u}{\partial x} \right) + y \left(\frac{\partial u}{\partial y} \right) + z \left(\frac{\partial u}{\partial z} \right) = 2$	(2017,2015)
9.	State Euler's theorem on homogenous function $u = x^n F\left(\frac{y}{x}\right)$ then show that $x \left(\frac{\partial u}{\partial x} \right) + y \left(\frac{\partial u}{\partial y} \right) = nu$	
10.	If $u = \log(x^3+y^3+z^3-3xyz)$ then show that $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \right)^2 u = \frac{9}{(x+y+z)^2}$	
11.	If u is a homogenous function of degree 'n' in x, y . then show that, $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = n(n-1)u$	

CHAPTER 8

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MAXIMUM AND MINIMUM VALUE, MEAN VALUE THEOREM

1.	Define Maximum And Minimum Value of a function.	
2.	Define extreme value.	
3.	Examine the function $f(x) = e^x + 2\cos x + e^{-x}$ for minimum values.	(2017)
4.	Prove that x^x has a minimum value for $x = \frac{1}{e}$	(2016,2012)
5.	Prove that $y = \sqrt{3} \sin x + 3\cos x$ has a maximum value at $x = \frac{\pi}{6}$	(2017, 2014, 2011)
6.	Determine the minimum and maximum value of the function $f(x) = x + \sin 2x$ for $0 < x < 2\pi$	(DCC-18)
7.	State and prove Rolle's theorem.	(2013, 2015, 2017)
8.	Verify the mean value theorem for the function $f(x) = 2 + 2x - x^2$ in the interval $(0,1)$	
9.	Verify the mean value theorem for the function $f(x) = x - x^3$ in the interval $(-2,1)$	
10.	Find the minimum and maximum value of function: $f(x) = x^5 - 5x^4 + 5x^3 - 1.$	
11.	State and prove Cauchy's mean value theorem.	2018
12.	State and prove Lagrange's Mean value theorem.	
13.	State Leibnitz's theorem.	2019
14.	State and prove Rolle's theorem.	2019

CHAPTER 9
CURVATURE

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1.	Find the radius and centre of the curvature at (4,4) of the curve $xy = 16$	(2012)
2.	Find the radius of the curvature of the curve $x^{\frac{2}{3}} + y^{\frac{2}{3}} = 1$ at $(\frac{1}{2\sqrt{2}}, \frac{1}{2\sqrt{2}})$	
3.	Find the radius of the curvature at the point (0,b) of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$	(2017)
4.	Find the length of the tangent and normal of $xy + 2x - y = 5$ at the point (2,1)	
5.	Find the equation of the tangent at the point 't' on the curve $x = a(t + \sin t), y = a(t - \cos t)$.	
6.	Prove that the portion of the tangent of the curve $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$ intercepted between the axes of its constant length.	(DCC-18)
7.	Find the center of curvature of the curve $xy=16$ at the point (4,4).	

INTEGRAL CALCULUS
CHAPTER 1
INDEFINITE INTEGRAL

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1.	$\int (x^a + a^x) dx$	
2.	$\int \frac{dx}{\sqrt{x} - \sqrt{(x-1)}}$	
3.	$\int \frac{dx}{\sqrt{x} + \sqrt{(x+1)}}$	
4.	$\int \frac{dx}{\sqrt{x} + \sqrt{(x-1)}}$	
5.	$\int \frac{dx}{1 + \sin x}$	
6.	$\int \frac{\sin^6 x + \cos^6 x}{\sin^2 x \cos^2 x} dx$	
7.	$\int \frac{\sin 2x}{\sin 5x \sin 3x} dx$	
8.	$\int \frac{\sin x dx}{\sin(x-a)}$	
9.	$\int \frac{dx}{\sin(x-a)\sin(x-b)}$	
10.	$\int \frac{2 - \sin 2x}{1 - \cos 2x} dx$	
11.	$\int \frac{\sin x dx}{\sqrt{1 + \cos x}}$	
12.	$\int \frac{\sin x dx}{\sqrt{(1 + \cos x)}}$	
13.	$\int \frac{\sin^3 x}{\cos^9 x}$	[2018]
14.	Evaluate : I. $\int \sin^5 x \cos^2 x dx$ II. $\int e^x \frac{(2 + \sin 2x)}{1 + \cos 2x} dx$ III. $\int \frac{7x-9}{x^2-2x+35} dx$ IV. $\int \frac{dx}{(x^2+4x+5)^2}$ V. $\int \frac{dx}{5+4\cos x}$ VI. $\int \frac{e^{m \tan^{-1} x}}{1+x^2} dx$	[2019]

CHAPTER 2 METHOD OF SUBSTITUTION

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1.	$\int \frac{(\tan x)^{\frac{3}{2}}}{\cos^2 x} dx$	(2016,2014,2012)
2.	$\int \frac{(1+x)e^x}{\cos^2(xe^x)} dx;$	(2017,2014,2013)
3.	$\int \frac{dx}{\sqrt[3]{x^3} \sqrt{(1+x)^5}}$	
4.	$\int \sqrt{1 + \sec x} dx$	(2015,2013)
5.	$\int \cos 2 \cot^{-1} \sqrt{\frac{1-x}{x+1}} dx$	
6.	$\int \frac{e^{a \sin^{-1} x}}{\sqrt{1-x^2}} dx$	
7.	$\int \frac{e^{m \tan^{-1} x}}{1+x^2} dx$	
8.	$\int x 7^{x^2} dx$	
9.	$\int \frac{dx}{x \sqrt{x^2-1}}$	
10.	$\int \frac{x^2+1}{x^4+1} dx$	
11.	$\int_0^{16} \frac{x^4}{1+\sqrt{x}} dx$	[2018]
12.	$\int \frac{dx}{(1+x)^2 \sqrt{1+x^2}}$	[2018]
13.	$\int_0^{\frac{\pi}{2}} \frac{\sqrt{\sin x}}{\sqrt{\sin x} - \sqrt{\cos x}} dx$	[2018][2019]
14.	$\int_0^{\pi} \frac{x \sin x}{1+\cos^2 x} dx$	[2018]
15.	$\int_0^{\frac{\pi}{4}} \ln(1+\tan \theta) d\theta$	[2018]
16.	Evaluate:	[2019]
	(i) $\int_0^{\frac{\pi}{2}} \frac{\sqrt{\sin x} dx}{\sqrt{\sin x} + \sqrt{\cos x}}$	
	(ii) $\int_0^{\infty} \frac{(\tan^{-1} x)^2}{1+x^2} dx$	
	(iii) $\int_0^{\frac{\pi}{2}} \frac{dx}{1+\cot x}$	
	(iv) $\int_0^{\frac{\pi}{2}} \sin^6 x \cos^3 x dx$	

17.	7.(doc file no.10).If $\ln = \int_0^{\frac{\pi}{4}} \tan \theta \, d\theta$ then show that, $\ln = \ln \sqrt{2}$	[2018]
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INTEGRATION BY PARTS

1.	$\int \sin^{-1} \sqrt{\frac{x}{a+x}} \, dx$	(2017, 2016, 2013)
2.	$\int \frac{e^{m \tan^{-1} x}}{(1+x^2)^2} \, dx$	(2017, 2014)
3.	$\int \cos \sqrt{x} \, dx$	
4.	$\int \frac{\ln(x+1)}{\sqrt{x+1}} \, dx$	
5.	$\int \cos^{-1} \left(\frac{1-x^2}{1+x^2} \right) \, dx$	
6.	$\int \frac{x \sin^{-1} x}{\sqrt{1-x^2}} \, dx$	
7.	$\int e^x \frac{1+\sin x}{1+\cos x} \, dx$	
8.	$\int \frac{x+\sin x}{1+\cos x} \, dx$	
9.	$\int \sqrt{25-9x^2} \, dx;$	
10.	$\int \frac{dx}{(x^2-1)\sqrt{x^2+1}};$	(DCC-18)
11.	$\int \sqrt{4+3x-2x^2} \, dx;$	

CHAPTER 4

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GAMMA-BETA FUNCTION

1.	Define gamma function	(2017,2015)
2.	Define beta function	(2017,2015)
3.	Prove that, if $a > 0$ then $i) \int_0^{\infty} e^{-at} t^{x-1} dt = \frac{\Gamma(x)}{a^x}, x > 0$ $ii) \int_0^{\infty} e^{-t^{\frac{1}{x}}} dt = \Gamma(x+1)$	
4.	If $x, y > 0$ then prove that, $\beta(x, y) = \beta(y, x)$	
5.	Define gamma function. Prove that $\int_0^{\frac{\pi}{2}} x \cos^n x dx = \frac{r(\frac{m+1}{2}) r(\frac{n+1}{2})}{2r(\frac{m+n+2}{2})}$	[2019]

CHAPTER 5

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DEFINITE INTEGRAL AS THE LIMIT OF SUM

1.	State and prove the fundamental theorem of integral calculus	(2016, 2014, 2012, 2011)
2.	Find the value of Find the value of $\int_a^b \sin x dx$ as a limit of sum as a limit of sum	(2017, 2013)
3.	Find the value of $\lim_{n \rightarrow \infty} \left[\frac{1}{n} + \frac{n^2}{(n+1)^3} + \frac{n^2}{(n+2)^3} + \dots + \frac{1}{8^n} \right]$	(2014, 2012)
4.	Find the value of $\lim_{n \rightarrow \infty} \left[\left(1 + \frac{1}{n}\right) \left(1 + \frac{2}{n}\right) \left(1 + \frac{3}{n}\right) \dots \left(1 + \frac{n}{n}\right) \right]^{\frac{1}{n}}$	
5.	Find the value of $\lim_{n \rightarrow \infty} \left\{ \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{n+n} \right\}$	
6.	Find the value of $\lim_{n \rightarrow \infty} \left[\frac{n^2}{n^3} + \frac{n^2}{(n+1)^3} + \frac{n^2}{(n+2)^3} + \dots + \frac{n^2}{(n+n)^3} \right]$	(2017, DCC-18)
7.	Find the value of $\lim_{n \rightarrow \infty} \left[\left(2 + \frac{1}{n^2}\right)^{\frac{1}{n^2}} \left(2 + \frac{2^2}{n^2}\right)^{\frac{2}{n^2}} \left(2 + \frac{3^2}{n^2}\right)^{\frac{3}{n^2}} \dots \left(2 + \frac{n^2}{n^2}\right)^{\frac{n^2}{n^2}} \right]$	(2016,20111)

	$\left(\frac{n^2}{n^2}\right)^{\frac{n}{n^2}}]$	
8.	Find the value of $\lim_{n \rightarrow \infty} \frac{1}{n} [(n+1)(n+2)(n+3) \dots (n+n)]^{\frac{1}{n}}$	(2013)

CHAPTER 6

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AREA OF PLANE REGIONS, RECTIFICATION AND VOLUME

1.	Show that the area between the parabola $y^2 = 4x$ and the straight line $y = 2x - 4$ is 9.	(2016, 2013, 2011)
2.	Find the area between the parabola $y^2 = 4x$ and the straight line $y = x$	(2017, 2015)
3.	Find the area of the loop curve $a^2 y^2 = x^2(a^2 - x^2)$	(2014)
4.	Find the area of the hypocycloid $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$	
5.	Find the whole area bounded by the curve $a^2 y^2 = x^3(2a - x)$	(2012)
6.	Find the area of the hypocycloid $x^{\frac{2}{3}} + y^{\frac{2}{3}} = 4$	
7.	Find the length of the perimeter of the asteroid $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$	(2014 v)
8.	Find the length of the arc of the parabola $y^2 = 12x$ cut off by the latus rectum.	(2012)
9.	Find the length of the perimeter of the cardioide $r = 4(1 - \cos\theta)$	(2011) (DCC-18)
10.	Find the length of the perimeter of the cardioide $r = 8(1 - \cos\theta)$	
11.	Find the length of the perimeter of the cardioide $r = 8(1 + \cos\theta)$	(2017, 2015, 2013)
12.	Find the length of the perimeter of the cardioide $r = a(1 + \cos\theta)$	
13.	Proved that the volume of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ on major axis is $\frac{4\pi a b^2}{3}$	(2017)
14.	Show that the volume of a sphere of radius a is $\frac{4}{3}\pi a^3$	(2016)
15.	Find the volume of the solid bounded by the surface of revolution about x axis of the curve $x = a(\theta - \sin\theta)$ and $y = a(1 - \cos\theta)$ between $\theta=0$ and $\theta=2\pi$.	(2016, 2012)

VOLUME OF SURFACES AND SOLIDS OF REVOLUTION

1. Show that the curvature of the cycloid $x = a(\theta - \sin\theta)$, $y = a(1 - \cos\theta)$ lines on a similar cycloid. [2010]
2. Find the length of area of the parabola $y^2 = 16x$ between the vertex and one end of the latus rectum. [2019]
3. Show that area between the parabola $y^2 = 4x$ and the straight Line $y=2x-4$ is 9. [2019]
4. Find the volume of the solid generated by revolving the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ about axis. [2019]

COORDINATE GEOMETRY

1. Prove that the straight line $lx + my + n = 0$ is a normal to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ if $\frac{a^2}{l^2} + \frac{b^2}{m^2} = \frac{(a^2 - b^2)^2}{n^2}$.
2. Prove that two of the lines represented by the equation $ax^4 + bx^3y + cx^2y^2 + dxy^3 + ay^4 = 0$ will bisect angle between the order two if $c + 6a = 0$ and $b + d = 0$ (2014)
3. Show that a homogeneous quadratic equation always represent a pair of straight line.
4. if the two straight lines represented by $x^2(\tan^2 \varphi + \cos^2 \varphi) - 2xy \tan \varphi + y^2 \sin^2 \varphi = 0$ make angle α and β with the axis of x, then show that $\tan \alpha - \tan \beta = 2$.
5. show that the equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents two parallel lines if $\frac{a}{h} = \frac{h}{b} = \frac{g}{f}$ and that when these conditions are satisfied, the distance between them is $2 \sqrt{\frac{g^2 - ac}{a(a+b)}}$.
6. Find the radical axis of the circles $x^2 + y^2 + 2x + 3y - 7 = 0$ and $x^2 + y^2 - 2x - y + 1 = 0$.
7. Find the general equation of all circles any pairs of which have the same radical axis as the circles $x^2 + y^2 = 4$ and $x^2 + y^2 + 2x + 4y = 6$.

SOME IMPORTANT FORMULAS

Differentiation Formulas:

1. $\frac{d}{dx}(x) = 1$
2. $\frac{d}{dx}(ax) = a$
3. $\frac{d}{dx}(x^n) = nx^{n-1}$
4. $\frac{d}{dx}(\cos x) = -\sin x$
5. $\frac{d}{dx}(\sin x) = \cos x$
6. $\frac{d}{dx}(\tan x) = \sec^2 x$
7. $\frac{d}{dx}(\cot x) = -\csc^2 x$
8. $\frac{d}{dx}(\sec x) = \sec x \tan x$
9. $\frac{d}{dx}(\csc x) = -\csc x(\cot x)$
10. $\frac{d}{dx}(\ln x) = \frac{1}{x}$
11. $\frac{d}{dx}(e^x) = e^x$
12. $\frac{d}{dx}(a^x) = (\ln a)a^x$
13. $\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$
14. $\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$
15. $\frac{d}{dx}(\sec^{-1} x) = \frac{1}{|x|\sqrt{x^2-1}}$

Integration Formulas:

1. $\int 1 dx = x + C$
2. $\int a dx = ax + C$
3. $\int x^n dx = \frac{x^{n+1}}{n+1} + C, n \neq -1$
4. $\int \sin x dx = -\cos x + C$
5. $\int \cos x dx = \sin x + C$
6. $\int \sec^2 x dx = \tan x + C$
7. $\int \csc^2 x dx = -\cot x + C$
8. $\int \sec x(\tan x) dx = \sec x + C$
9. $\int \csc x(\cot x) dx = -\csc x + C$
10. $\int \frac{1}{x} dx = \ln |x| + C$
11. $\int e^x dx = e^x + C$
12. $\int a^x dx = \frac{a^x}{\ln a} + C, a > 0, a \neq 1$
13. $\int \frac{1}{\sqrt{1-x^2}} dx = \sin^{-1} x + C$
14. $\int \frac{1}{1+x^2} dx = \tan^{-1} x + C$
15. $\int \frac{1}{|x|\sqrt{x^2-1}} dx = \sec^{-1} x + C$

TRIGONOMETRIC FORMULA

$$(1) \quad 1 + \cos x = 2 \cos^2 (x/2)$$

(Equivalently, $1 + \cos 2x = 2 \cos^2 x$)

$$(2) \quad 1 - \cos x = 2 \cos^2 (x/2)$$

(Equivalently, $1 - \cos 2x = 2 \cos^2 x$)

$$(3) \quad \sin A \cos B = \frac{1}{2} [\sin (A+B) + \sin (A-B)]$$

$$(4) \quad \cos A \sin B = \frac{1}{2} [\sin (A+B) - \sin (A-B)]$$

$$(5) \quad \cos A \cos B = \frac{1}{2} [\cos (A+B) + \cos (A-B)]$$

$$(6) \quad \sin A \sin B = \frac{1}{2} [\cos (A-B) - \cos (A+B)]$$

$$\triangleright \sin (A+B) = \sin A \cos B + \cos A \sin B$$

$$\triangleright \sin (A-B) = \sin A \cos B - \cos A \sin B$$

$$\triangleright \cos (A+B) = \cos A \cos B - \sin A \sin B$$

$$\triangleright \cos (A-B) = \cos A \cos B + \sin A \sin B$$

$$\triangleright \tan (A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\triangleright \tan (A-B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

$$\triangleright \sin (A+B) \sin (A-B) = \sin^2 A - \sin^2 B = \cos^2 B - \cos^2 A$$

$$\triangleright \cos (A+B) \cos (A-B) = \cos^2 A - \sin^2 B = \cos^2 B - \sin^2 A$$

$$\triangleright \sin 2A = 2 \sin A \cos A = \frac{2 \tan A}{1 + \tan^2 A}$$

$$\triangleright \cos 2A = \cos^2 A - \sin^2 A = 1 - 2 \sin^2 A = 2 \cos^2 A - 1 = \frac{1 - \tan^2 A}{1 + \tan^2 A}$$

$$\triangleright \tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

$$\triangleright \sin 3A = 3 \sin A - 4 \sin^3 A = 4 \sin (60^\circ - A) \sin A \sin (60^\circ + A)$$

$$\triangleright \cos 3A = 4 \cos^3 A - 3 \cos A = 4 \cos (60^\circ - A) \cos A \cos (60^\circ + A)$$

$$\triangleright \tan 3A = \frac{3 \tan A - \tan^3 A}{1 - 3 \tan^2 A} = \tan (60^\circ - A) \tan A \tan (60^\circ + A)$$

SURFACE FORMULA

Name	Curved Surface Area	Total Surface Area	Volume
Cuboid	$2(l + b)h$	$2(lb + bh + hl)$	lbh
Cube	$4a^2$	$6a^2$	a^3
Right Circular Cylinder	$2\pi rh$	$2\pi r(h + r)$	$\pi r^2 h$
Right circular cone	πrl	$\pi r(l + r)$	$\frac{1}{3}\pi r^2 h$
Sphere	$4\pi r^2$	$4\pi r^2$	$\frac{4}{3}\pi r^3$
Solid Hemisphere	$2\pi r^2$	$3\pi r^2$	$\frac{2}{3}\pi r^3$
Hollow Hemisphere	$2\pi r^2$	$2\pi r^2$	$\frac{2}{3}\pi r^3$

This book is very helpful for NU CSE students !!



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